

all-sites-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

metadata <- read_csv(here("data", "metadata.csv"))

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc"))

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc")) |>
  filter(site != "coal_3")

coal_creek_all <- compiled_isotope |>
  filter(location %in% c("cc"))
```

```

italian_creek <- compiled_isotope |>
  filter(location == "ic")

# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"),
  show_col_types = FALSE) |> # reading in data
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,
    bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
    date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on
the day
    avg_ap = mean(ap, na.rm = TRUE),
    avg_temp = mean(temp, na.rm = TRUE),
    avg_water_level = mean(water_level, na.rm = TRUE),
    avg_bp = mean(bp, na.rm = TRUE),
    total_readings = n(), # the amount of data it used to average
    .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types =
FALSE) |> # reading in data
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,
    bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
    date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on
the day

```

```

      avg_ap = mean(ap, na.rm = TRUE),
      avg_temp = mean(temp, na.rm = TRUE),
      avg_water_level = mean(water_level, na.rm = TRUE),
      avg_bp = mean(bp, na.rm = TRUE),
      total_readings = n(), # the amount of data it used to average
      .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_data,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)

tc_isco <- isco_data |>
  filter(site %in% c("tc"))

utc_isco <- isco_data |>
  filter(site %in% c("utc"))

```

Statistical Tests

```

#----- ANOVA of lc-excess and location -----
#
anova <- aov(lc_excess ~ location, data = compiled_isotope)
summary(anova)

```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
location	3	190.5	63.51	31.68	1.06e-14 ***
Residuals	106	212.5	2.00		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
TukeyHSD(anova)
```

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = lc_excess ~ location, data = compiled_isotope)

```
$location
      diff      lwr      upr    p adj
it-cc  2.7056933  1.1233963  4.2879903 0.0001176
tc-cc -1.9272417 -3.1199429 -0.7345404 0.0003008
utc-cc -0.8415317 -2.0578531  0.3747897 0.2762889
tc-it  -4.6329350 -5.9175136 -3.3483564 0.0000000
utc-it -3.5472250 -4.8537638 -2.2406862 0.0000000
utc-tc  1.0857100  0.2945615  1.8768585 0.0028641
```

```
#----- Correlation od location and date
----- #

compiled_isotope$date_c <- as.numeric(compiled_isotope$date) -
mean(as.numeric(compiled_isotope$date))
# center the date variable by subtracting its mean.

model <- lmer(
  lc_excess ~ location + date_c + # fixed effects: location differences + linear
time trend
  (1|site), # random intercept: allows each site to have its own baseline lc-
excess, accounting for
           # the ~36% of variance that's due to site-level clustering rather
than location or date
  data = compiled_isotope)

summary(model)
```

Linear mixed model fit by REML ['lmerMod']
Formula: lc_excess ~ location + date_c + (1 | site)
Data: compiled_isotope

REML criterion at convergence: 316

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.16719	-0.52062	-0.01911	0.69356	1.93212

Random effects:

Groups	Name	Variance	Std.Dev.
site	(Intercept)	0.4606	0.6787
Residual		0.8236	0.9075

Number of obs: 110, groups: site, 20

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	0.425745	0.394113	1.080
locationit	3.120881	0.566898	5.505
locationtc	-2.983319	0.529035	-5.639
locationutc	-1.671252	0.557774	-2.996
date_c	-0.048846	0.004535	-10.771

Correlation of Fixed Effects:

	(Intr)	loctnt	locationtc	locationutc
locationit	-0.634			
locationtc	-0.715	0.480		
locationutc	-0.685	0.454	0.499	
date_c	-0.253	-0.068	0.068	0.093

centering doesn't affect slopes, only the intercept's ^ from the (1|site)

anova(model)

Analysis of Variance Table

	npair	Sum Sq	Mean Sq	F value
location	3	78.793	26.264	31.891
date_c	1	95.543	95.543	116.012

#----- Correlation with normalized stream distance
----- #

Trail Creek 6/24 - lc-excess & cumulative dams

```
model_6_24_str_tc <- lm(`lc-excess_6-24` ~ normalized_stream_distance_m, data  
= gradient_tc)  
summary(model_6_24_str_tc)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ normalized_stream_distance_m,
```

```

data = gradient_tc)

Residuals:
      1      2      3      4      5      6      7
-0.36861 -0.32358  0.58444  0.03641  0.15093  0.31789 -0.39748

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -1.3907116   0.2562773   -5.427 0.002880 **
normalized_stream_distance_m -0.0011403   0.0001537  -7.417 0.000702 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.416 on 5 degrees of freedom
Multiple R-squared:  0.9167,    Adjusted R-squared:    0.9
F-statistic: 55.01 on 1 and 5 DF,  p-value: 0.0007016

```

```

# Trail Creek 6/30 - lc-excess & cumulative dams
model_6_30_str_tc <- lm(`lc-excess_6-30` ~ normalized_stream_distance_m, data
= gradient_tc)
summary(model_6_30_str_tc)

```

```

Call:
lm(formula = `lc-excess_6-30` ~ normalized_stream_distance_m,
    data = gradient_tc)

Residuals:
      1      2      3      4      5      6      7
-0.08528 -0.20669  0.24371  0.25137  0.19568 -0.61753  0.21874

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2.2800413   0.2204137  -10.34 0.000145 ***
normalized_stream_distance_m -0.0008555   0.0001322   -6.47 0.001314 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3578 on 5 degrees of freedom
Multiple R-squared:  0.8933,    Adjusted R-squared:    0.872
F-statistic: 41.86 on 1 and 5 DF,  p-value: 0.001314

```

```
# Trail Creek 7/15 - lc-excess & cumulative dams
model_7_15_str_tc <- lm(`lc-excess_7-15` ~ normalized_stream_distance_m, data
= gradient_tc)
summary(model_7_15_str_tc)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

1	2	3	4	5	6	7
-0.36655	-0.34573	0.61325	0.47415	-0.07123	-0.28328	-0.02060

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.1034551	0.2671852	-4.130	0.00909 **
normalized_stream_distance_m	-0.0010277	0.0001603	-6.412	0.00137 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4337 on 5 degrees of freedom

Multiple R-squared: 0.8916, Adjusted R-squared: 0.8699

F-statistic: 41.11 on 1 and 5 DF, p-value: 0.001369

```
# _____ #
```

```
# Coal Creek 6/23 - lc-excess & cumulative dams
model_6_23_str_cc <- lm(`lc-excess_6-23` ~ normalized_stream_distance_m, data
= gradient_cc)
summary(model_6_23_str_cc)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

1	2	3	4	5
-0.3323	0.6279	0.2566	-0.3412	-0.2109

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.5393266	0.4492095	3.427	0.0416 *
normalized_stream_distance_m	-0.0006445	0.0002206	-2.921	0.0614 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4938 on 3 degrees of freedom

Multiple R-squared: 0.7399, Adjusted R-squared: 0.6532

F-statistic: 8.533 on 1 and 3 DF, p-value: 0.06144

```
# # Coal Creek 7/7 - lc-excess & cumulative dams
model_7_7_str_cc <- lm(`lc-excess_7-7` ~ normalized_stream_distance_m, data
= gradient_cc)
summary(model_7_7_str_cc)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

1	2	3	4	5
0.2092	0.3569	-0.8765	-0.8080	1.1184

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.333192	0.885667	-0.376	0.7318
normalized_stream_distance_m	-0.001031	0.000435	-2.369	0.0986 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9735 on 3 degrees of freedom

Multiple R-squared: 0.6517, Adjusted R-squared: 0.5356

F-statistic: 5.613 on 1 and 3 DF, p-value: 0.09859

```
#----- Correlation of Trail Creek Upstream and Downstream
----- #
cor.test(tc_isco$avg_water_level, tc_isco$lc_excess)
```


Pearson's product-moment correlation

```
data: tc_isco$avg_water_level and tc_isco$lc_excess
t = 12.003, df = 46, p-value = 8.995e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7792542 0.9257501
sample estimates:
      cor
0.8706218
```

```
cor.test(utc_isco$avg_water_level, utc_isco$lc_excess)
```

Pearson's product-moment correlation

```
data: utc_isco$avg_water_level and utc_isco$lc_excess
t = 4.7707, df = 38, p-value = 2.71e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3713137 0.7756583
sample estimates:
      cor
0.6120308
```

```
#----- Correlation with cumulative dams -----
#
gradient_tc$cumulative_dams
<- as.numeric(as.character(gradient_tc$cumulative_dams))

# Trail Creek 6/24 - lc-excess & cumulative dams
model_6_24_cum_tc <- lm(`lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
summary(model_6_24_cum_tc)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

```
      2      3      4      5      6      7
```

```
-0.4806 0.1654 0.1151 0.4332 0.4826 -0.7157
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.613391	0.362589	-4.450	0.01125 *
cumulative_dams	-0.034669	0.006788	-5.107	0.00695 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5487 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.867, Adjusted R-squared: 0.8338

F-statistic: 26.08 on 1 and 4 DF, p-value: 0.006948

```
# Trail Creek 6/30 - lc-excess & cumulative dams
```

```
model_6_30_cum_tc <- lm(`lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
summary(model_6_30_cum_tc)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

2	3	4	5	6	7
-0.27039	-0.01815	0.34655	0.43245	-0.47798	-0.01249

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.501209	0.257870	-9.699	0.000632 ***
cumulative_dams	-0.025481	0.004828	-5.278	0.006179 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3902 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8744, Adjusted R-squared: 0.843

F-statistic: 27.86 on 1 and 4 DF, p-value: 0.006179

```
# Trail Creek 7/15 - lc-excess & cumulative dams
```

```
model_7_15_cum_tc <- lm(`lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
summary(model_7_15_cum_tc)
```

```

Call:
lm(formula = `lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)

Residuals:
      2      3      4      5      6      7
-0.55012  0.17680  0.52850  0.19513 -0.09976 -0.25056

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.241283   0.281127  -4.415  0.01155 *
cumulative_dams -0.032606   0.005263  -6.195  0.00345 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4254 on 4 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.9056,    Adjusted R-squared:  0.882
F-statistic: 38.38 on 1 and 4 DF,  p-value: 0.003452

```

```

# _____ #

gradient_cc$cumulative_dams
<- as.numeric(as.character(gradient_cc$cumulative_dams))

# Coal Creek 6/23 - lc-excess & cumulative dams
model_6_23_cum_cc <- lm(`lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
summary(model_6_23_cum_cc)

```

```

Call:
lm(formula = `lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)

Residuals:
      2      3      4      5
0.11851  0.07466 -0.40553  0.21236

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.69377   0.41827   4.049  0.0559 .
cumulative_dams -0.04687   0.01195  -3.921  0.0593 .
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3385 on 2 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.8849, Adjusted R-squared: 0.8273
F-statistic: 15.37 on 1 and 2 DF, p-value: 0.05931

```
# # Coal Creek 7/7 - lc-excess & cumulative dams
model_7_7_cum_cc <- lm(`lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
summary(model_7_7_cum_cc)
```

Call:
lm(formula = `lc-excess\_7-7` ~ cumulative_dams, data = gradient_cc)

Residuals:

2	3	4	5
0.8361	-0.6905	-0.9591	0.8135

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.13831	1.45089	-1.474	0.278
cumulative_dams	-0.01660	0.04146	-0.400	0.728

Residual standard error: 1.174 on 2 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.0742, Adjusted R-squared: -0.3887
F-statistic: 0.1603 on 1 and 2 DF, p-value: 0.7276

```
#----- Correlation with cumulative dam length
----- #
gradient_tc$cumulative_dam_length
<- as.numeric(as.character(gradient_tc$cumulative_dam_length))

# Trail Creek 6/24 - lc-excess & cumulative dam length
model_6_24_cumd_tc <- lm(`lc-excess_6-24` ~ cumulative_dam_length, data =
gradient_tc)
summary(model_6_24_cumd_tc)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.3782	0.3121	-0.2486	0.3477	0.3552	-0.3882

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.715833	0.259857	-6.603	0.00273 **
cumulative_dam_length	-0.056153	0.008125	-6.911	0.00230 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4183 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9227, Adjusted R-squared: 0.9034

F-statistic: 47.77 on 1 and 4 DF, p-value: 0.002299

```
# Trail Creek 6/30 - lc-excess & cumulative dam length
model_6_30_cumd_tc <- lm(`lc-excess_6-30` ~ cumulative_dam_length, data =
gradient_tc)
summary(model_6_30_cumd_tc)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.16963	0.11233	0.09129	0.36273	-0.58878	0.19207

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.601968	0.233442	-11.15	0.000369 ***
cumulative_dam_length	-0.040214	0.007299	-5.51	0.005295 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3758 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8836, Adjusted R-squared: 0.8545

F-statistic: 30.36 on 1 and 4 DF, p-value: 0.005295

```
# Trail Creek 7/15 - lc-excess & cumulative dam length
model_7_15_cumd_tc <- lm(`lc-excess_7-15` ~ cumulative_dam_length, data =
gradient_tc)
summary(model_7_15_cumd_tc)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

```
      2      3      4      5      6      7
-0.4429  0.3245  0.1916  0.1118 -0.2269  0.0420
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.348532	0.197391	-6.832	0.00240 **
cumulative_dam_length	-0.052360	0.006172	-8.484	0.00106 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3178 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9474, Adjusted R-squared: 0.9342

F-statistic: 71.97 on 1 and 4 DF, p-value: 0.001058

```
# _____ #
```

```
gradient_cc$cumulative_dam_length
<- as.numeric(as.character(gradient_cc$cumulative_dam_length))
```

```
# Coal Creek 6/23 - lc-excess & cumulative dam length
```

```
model_6_23_cumd_cc <- lm(`lc-excess_6-23` ~ cumulative_dam_length, data =
gradient_cc)
summary(model_6_23_cumd_cc)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

```
      2      3      4      5
```

```
0.10405 -0.06714 -0.20308 0.16617
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.547568	0.225772	6.855	0.0206 *
cumulative_dam_length	-0.027556	0.004094	-6.730	0.0214 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2052 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9577, Adjusted R-squared: 0.9366

F-statistic: 45.29 on 1 and 2 DF, p-value: 0.02137

```
# Coal Creek 7/7 - lc-excess & cumulative dam length
model_7_7_cumd_cc <- lm(`lc-excess_7-7` ~ cumulative_dam_length, data =
gradient_cc)
summary(model_7_7_cumd_cc)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

2	3	4	5
0.7792	-0.7622	-0.8651	0.8481

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.11270	1.26808	-1.666	0.238
cumulative_dam_length	-0.01134	0.02300	-0.493	0.671

Residual standard error: 1.152 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.1083, Adjusted R-squared: -0.3375

F-statistic: 0.243 on 1 and 2 DF, p-value: 0.6709

```
#----- Correlation with average water level
----- #
```

```
# Trail Creek - lc-excess & average water level
model_water_tc <- lm(lc_excess ~ avg_water_level, data = tc_isco)
summary(model_water_tc)
```

Call:

```
lm(formula = lc_excess ~ avg_water_level, data = tc_isco)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.48002	-0.40921	-0.06903	0.29693	2.43754

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.0594	0.3926	-17.98	<2e-16 ***
avg_water_level	14.2590	1.1880	12.00	9e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8348 on 46 degrees of freedom

Multiple R-squared: 0.758, Adjusted R-squared: 0.7527

F-statistic: 144.1 on 1 and 46 DF, p-value: 8.995e-16

```
# Upstream Trail Creek (has no beaver dams)- lc-excess & average water level
model_water_utc <- lm(lc_excess ~ avg_water_level, data = utc_isco)
summary(model_water_utc)
```

Call:

```
lm(formula = lc_excess ~ avg_water_level, data = utc_isco)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.4145	-0.5655	0.0331	0.6024	1.2292

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.800	1.328	-5.875	8.45e-07 ***
avg_water_level	15.757	3.303	4.771	2.71e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7075 on 38 degrees of freedom
Multiple R-squared: 0.3746, Adjusted R-squared: 0.3581
F-statistic: 22.76 on 1 and 38 DF, p-value: 2.71e-05

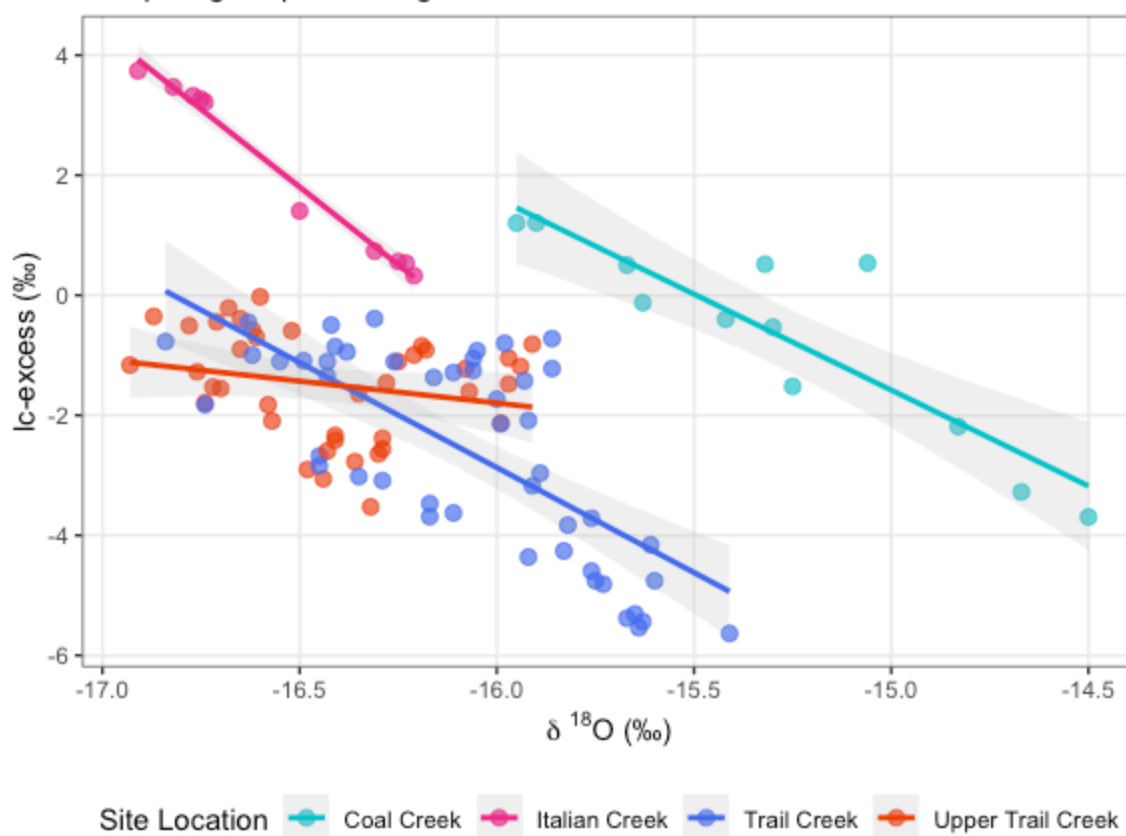
Exploratory Visuals with ALL Sites

Line-Conditioned Excess (lc-excess) vs Delta 18O

```
ggplot(data = compiled_isotope, aes(x = delta_o, y = lc_excess, color = location))  
+  
  geom_point(size = 2.5, alpha = 0.7) +  
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +  
  scale_color_manual(  
    values = c("tc" = "royalblue2", "cc" = "turquoise3", "it" = "deeppink2",  
    "utc" = "orangered2"),  
    labels = c("cc" = "Coal Creek", "it" = "Italian Creek", "tc" = "Trail Creek",  
    "utc" = "Upper Trail Creek")  
  ) +  
  labs(  
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",  
    subtitle = "Comparing evaporation signals between sites",  
    x = expression(paste(delta, " ^18, "O (\u2030))), # Formats as  $\delta^{18}\text{O}$  (‰)  
    y = "lc-excess (\u2030)",  
    color = "Site Location") +  
  theme_bw() +  
  theme(plot.title = element_text(face = "bold", size = 14),  
    legend.position = "bottom",  
    panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
  aes(x = date,
    y = lc_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(linewidth = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
```

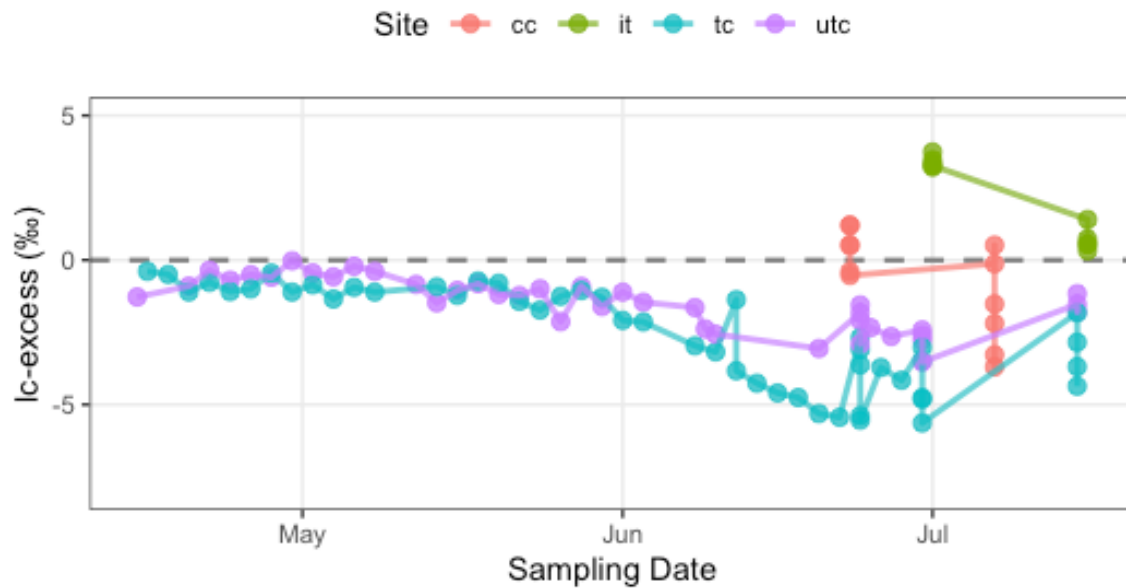
```

x = "Sampling Date",
y = "lc-excess (\u2030)",
color = "Site") +
ylim(-8,5) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")

```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment



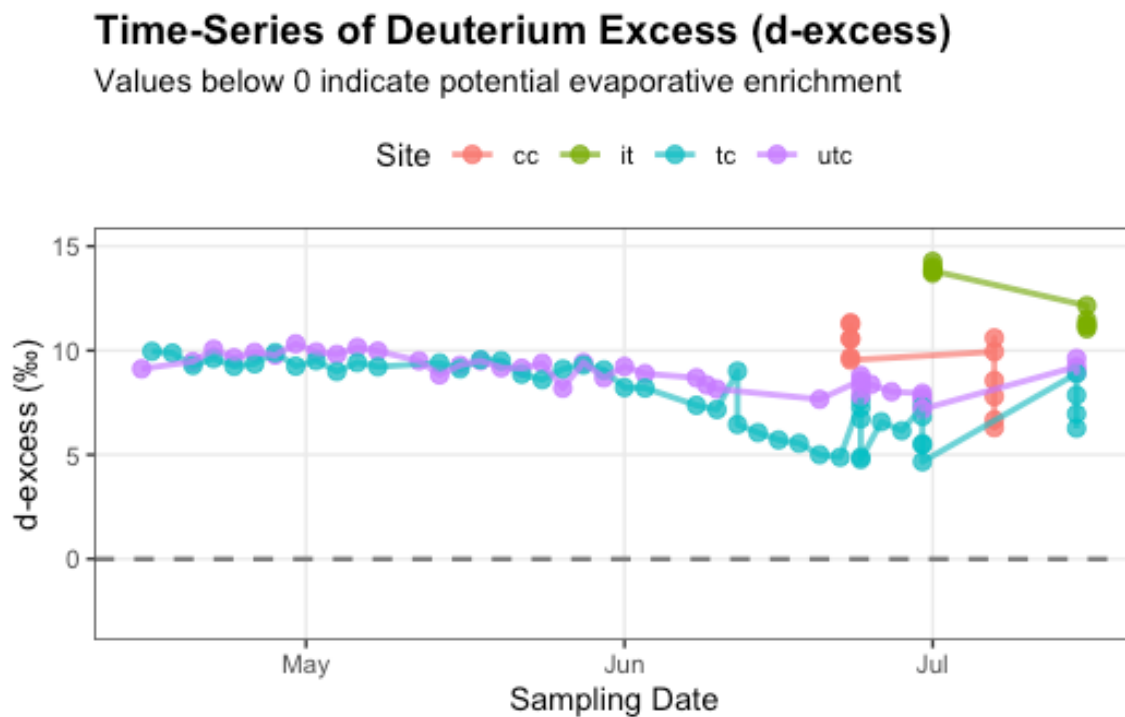
Time-Series of Deuterium Excess (d-excess)

```

ggplot(data = compiled_isotope,
       aes(x = date,
           y = d_excess,
           color = location,
           group = location)) +
geom_hline(yintercept = 0,
           linetype = "dashed",
           color = "gray50",
           size = 0.8) +
geom_point(size = 2.5,
           alpha = 0.8) +

```

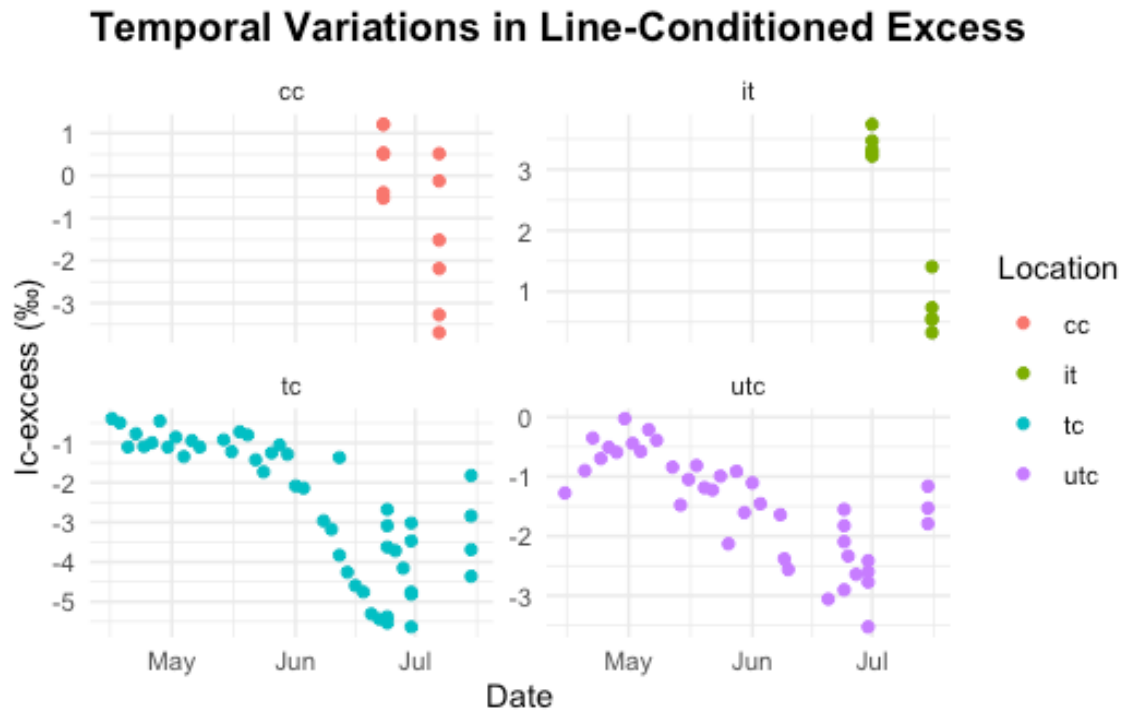
```
geom_line(size = 1,
          alpha = 0.7) +
labs(title = "Time-Series of Deuterium Excess (d-excess)",
     subtitle = "Values below 0 indicate potential evaporative enrichment",
     x = "Sampling Date",
     y = "d-excess (‰)",
     color = "Site") +
ylim(-3,15) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess (by location)

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = location)) +
geom_point(size = 1.5) +
labs(
  title = "Temporal Variations in Line-Conditioned Excess",
  x = "Date",
  y = "lc-excess (‰)",
  color = "Location")
```

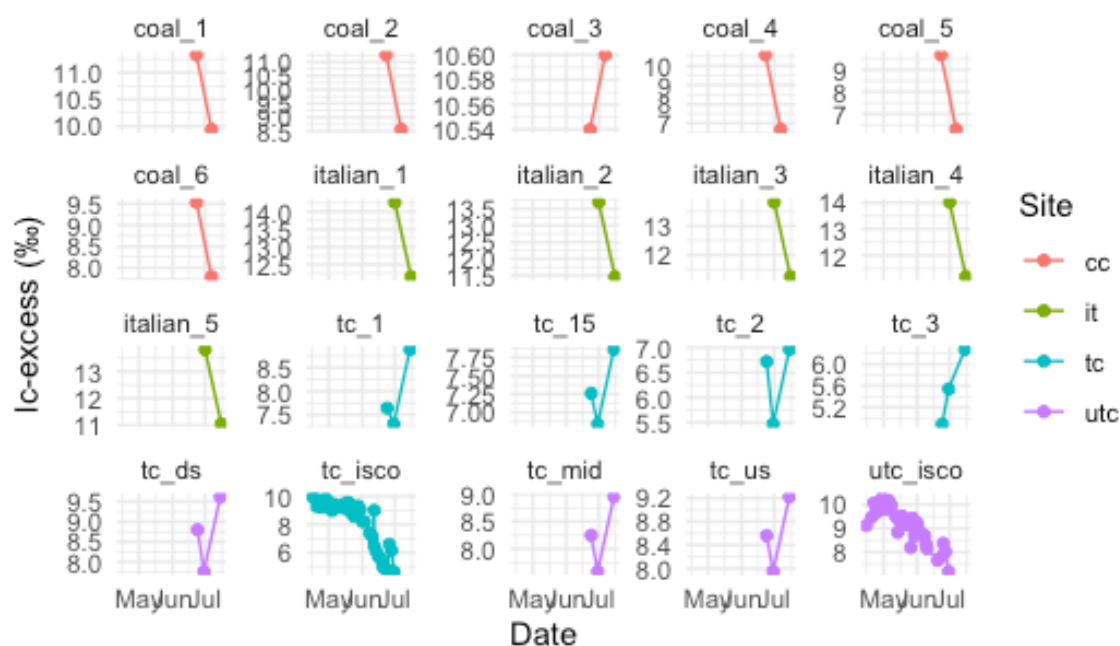
```
) +
facet_wrap(~location, scales = "free_y") + # Compare across sites easily
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))
```



Temporal Variations in Deuterium Excess (by sites)

```
ggplot(compiled_isotope, aes(x = date, y = d_excess, color = location)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Deuterium Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Deuterium Excess

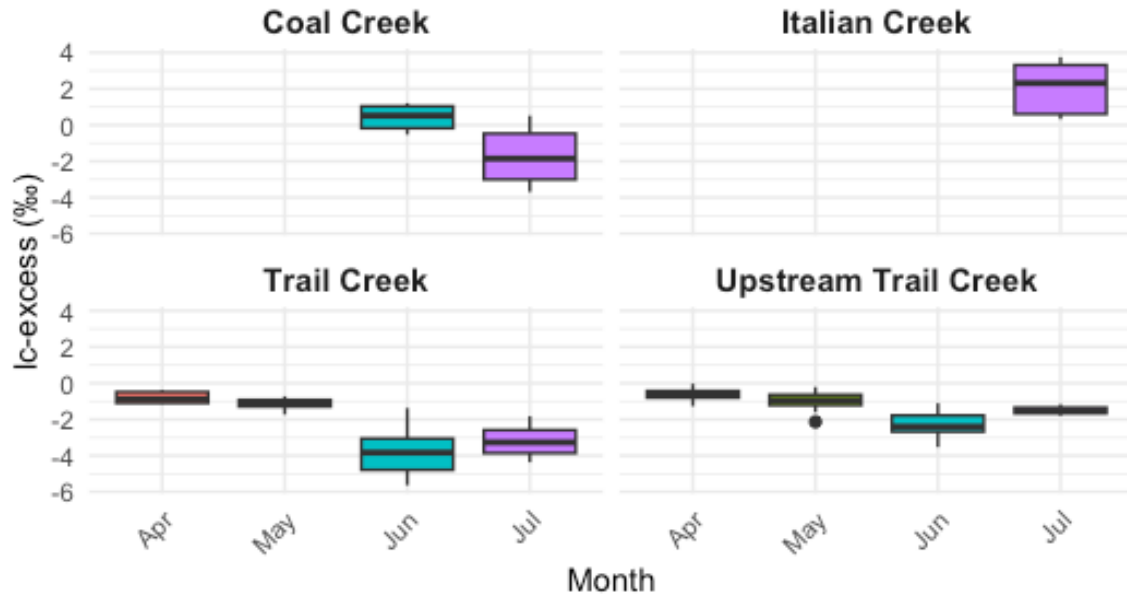


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(compiled_isotope, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
cc = "Coal Creek",
it = "Italian Creek",
utc = "Upstream Trail
Creek")))) +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1)) # tilting the x axis text
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



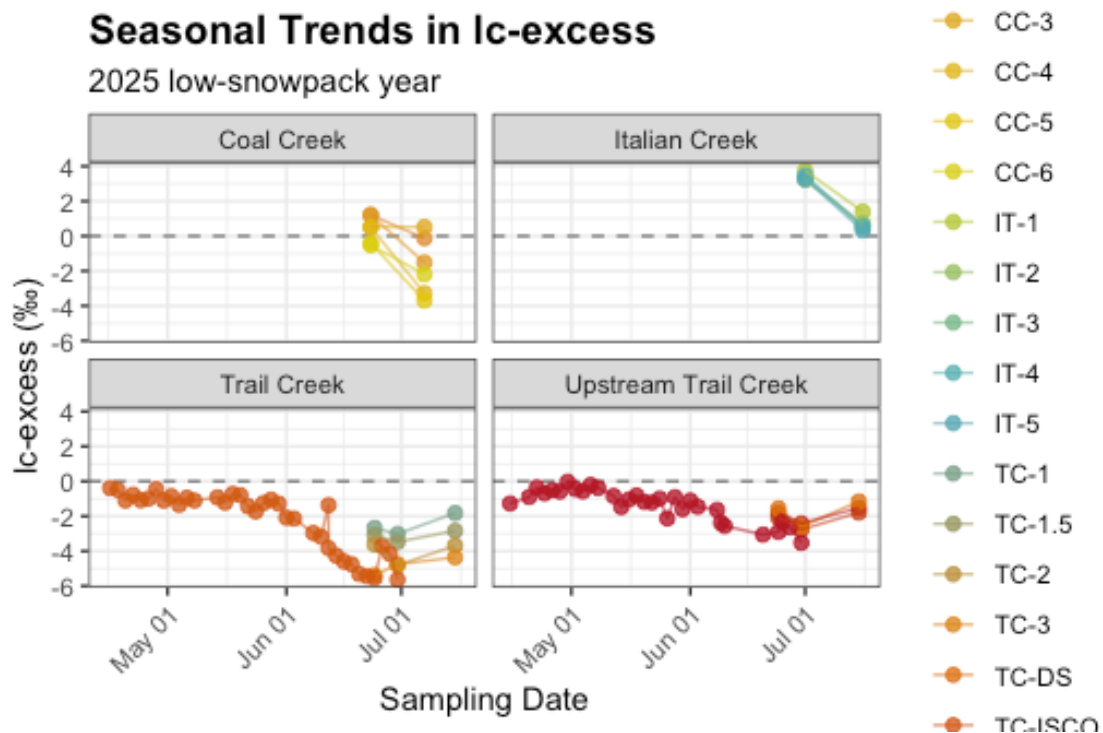
Time series of lc-excess across season, colored by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         it = "Italian Creek",
                                                         utc = "Upstream Trail
Creek")))) +
  scale_x_date(date_labels = "%b %d") +
  scale_color_manual(values = wes_palette("FantasticFox1", n =
length(unique(compiled_isotope$site)),
                                                         type = "continuous"),
labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3"
= "CC-3", "coal_4" =
"CC-4", "coal_5" = "CC-5", "coal_6" = "CC-6",
"italian_1" = "IT-1",
"italian_2" = "IT-2", "italian_3" = "IT-3",
"italian_4" = "IT-4",
"italian_5" = "IT-5", "tc_isco" = "TC-ISCO",
```

```

"tc_1" = "TC-1", "tc_15"
                                = "TC-1.5", "tc_2" = "TC-2", "tc_3" = "TC-3",
"tc_ds" = "TC-DS",
                                "tc_mid" = "TC-Mid", "tc_us" = "TC-US", "utc_isco"
= "UTC-ISCO")) +
  labs(title = "Seasonal Trends in lc-excess",
        subtitle = "2025 low-snowpack year",
        x = "Sampling Date", y = "lc-excess (‰)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        axis.text.x = element_text(angle = 45, hjust = 1))

```



Dual-isotope plot ($\delta^{18}\text{O}$ vs δD) with LMWL reference

```

# Add Global Meteoric Water Line (GMWL:  $y = 8x + 10$ )

gmwl_slope <- 8.00
gmwl_intercept <- 10.00

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +

```



```

geom_abline(slope = gmw1_slope, intercept = gmw1_intercept,
            linetype = "dashed", color = "black", linewidth = 0.8) +
facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                    cc = "Coal Creek",
                                                    it = "Italian Creek")) +
                                                    utc = "Upstream
Trail Creek",

labs(title = "Dual Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
     subtitle = "Dashed line = Global Meteoric Water Line",
     x = expression(paste(delta^18, "O (‰)")),
     y = expression(paste(delta, "D (‰)")),
     color = "Reach") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      legend.position = "none")

```

